

Nonnegative and positive factorizations

Full nonnegative rank of the quadratic correlation matrix

Difficulty: extreme **Importance:** broadly interesting

Status: Solved

Rating rationale: Historical ratings for the original conjecture. Extreme because full nonnegative rank is much sharper than available exponential lower bounds for this family; broad importance concerns communication complexity and limitations of linear programming formulations.

Area: exact NMF and lower bounds for optimization formulations

Negative resolution — 2026-09-13

Author: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation. **Independent Codex AI-agent informal audit:** PASS for the complete original target.

Theorem 1, proved in Sections 2–4 constructs nonnegative rational factors for the fully prescribed matrix and proves

$$\text{rank}_+(C_n) \leq \min \left\{ 2^n, 2^{n-1} + n + \binom{n}{2} + \binom{n}{4} \right\}, \quad n \geq 1.$$

In particular, $\text{rank}_+(C_7) \leq 127 < 128$, so the original equality for every $n \geq 3$ is false. The bound is strictly below 2^n for every $n \geq 7$. All entries remain fixed by the original formula, over the original real field; positive column denominators convert the exact integer certificate into genuine rational factors. This is a complete negative resolution of the universal question. Exact ranks at $n = 5, 6, 7$ and the smallest counterexample dimension are not determined or required, and no extension-complexity claim for the whole correlation polytope is made.

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The complete argument passed independent informal AI-agent review, with fresh exact checks. ChatGPT assistance is disclosed; neither external human peer review nor formal verification is claimed. No Lean verification was performed. The original statement and prior partial-result credit are retained below.

Historical partial result — 2026-09-11

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent Codex-agent review:** PASS for the stated partial scope.

The fixed three-bit quadratic correlation matrix has nonnegative rank exactly eight. Its ordinary-rank-seven parity null vector constrains both factors in a hypothetical seven-term factorization; nine distinguished entries then exclude such a factorization.

Remaining question at that date: The full prescribed-completion conjecture for every $n \geq 4$ remained unresolved. The parity restriction is proved only for a hypothetical factorization whose inner dimension equals ordinary rank; it is not imposed on arbitrary wider factorizations. The new counterexample above now settles the universal conjecture negatively; these earlier findings retain their original credit.

Primary reference: [complete authored PDF](#), [standalone TeX](#), [Theorem 1 and Lemma 2](#). [Independent proof review](#) · [Authorship and submission record](#). Verification is independent agent review, not external human peer review or formal certification.

Context and notation

All factorizations are over the real numbers. For $X \in \mathbb{R}_{\geq 0}^{m \times n}$, define

$$\text{rank}_+(X) = \min\{r \geq 0 : X = WH, \quad W \in \mathbb{R}_{\geq 0}^{m \times r}, H \in \mathbb{R}_{\geq 0}^{r \times n}\}.$$

Problem statement

For each integer $n \geq 3$, let C_n be the $2^n \times 2^n$ matrix indexed by $a, b \in \{0, 1\}^n$ and defined by

$$C_n(a, b) = (1 - a^\top b)^2.$$

Question

Is $\text{rank}_+(C_n) = 2^n$ for every $n \geq 3$? All entries, including those for $a^\top b > 1$, are fixed by this formula.

The matrix has ordinary rank $1 + n(n+1)/2$ and is a submatrix of a slack matrix of the correlation polytope formed from valid, possibly redundant inequalities. The conjecture therefore asks for a large separation between ordinary and nonnegative matrix rank with consequences for linear programming representations.

References

Vandaele, Gillis, Glineur, and Tuytens, *Heuristics for Exact Nonnegative Matrix Factorization*, §6.4, Conjecture 4. Gillis, *Nonnegative Matrix Factorization*, §3.7, p. 96, using the name U_n for this fixed matrix.

Status check — 2026-09-10

Rechecked [Baeckelant et al. v2, §6.7 and Appendix A.4](#) and [Sergeev's July 2026 paper](#), and searched for a full-rank resolution. The former retains the prescribed-completion conjecture, including the unresolved $n=3$ case. Sergeev treats a partial unique-disjointness matrix whose remaining entries may vary; this does not determine the rank of the fixed matrix here. No full resolution was located.